# David Reyes Alés

Software Engineer · Backend & Infrastructure · Applied AI

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## Profile

Software Engineer with 2.5+ years building backend services, real-time distributed systems and CI/CD infrastructure in production, now focused on bringing AI into reliable software. Backend experience across the JVM and Node.js ecosystems — Java, Spring Boot and NestJS — with Docker and SQL/NoSQL data stores. Applied ML/NLP with PyTorch and Transformers (paper published at IberLEF 2026) and rigorous evaluation of LLM-based systems.

## Work Experience

**Software Developer** — March 2026 - July 2026
*The Reuse Company · UC3M* — Madrid, Spain

Built a **real-time collaborative SysML v2 modelling tool** (industry research project, MSc thesis) in an Agile team: ASP.NET + PostgreSQL backend and HocusPocus (Yjs CRDTs) for **conflict-free multi-user editing**.

`React/NextJS` `ASP.NET` `PostgreSQL` `HocusPocus` `SysMLv2` `Agile`

**Full-Stack Developer** — 2023 – 2025, (last months alongside MSc studies)
*Axion · Telecommunications* — Spain

Joined a 3-person team to rebuild from scratch a nationwide real-time monitoring platform for broadcast infrastructure — a project that had failed over 1.5 years of prior attempts. Delivered in 6 months; adopted by 100+ operations users, after all previous versions had been rejected.

- **Performance:** Identified CouchDB (NoSQL) bottleneck, introduced **Redis Stack** as caching layer (validated via POC), reducing load times **50–60%** (10–11s → 4–5s). Also enabled HTTP compression at zero infrastructure cost.
- **CI/CD:** Built GitLab CI/CD pipeline from scratch on own initiative, cutting deployment time from **5–8h to 40–60 min** (85%+ reduction), with automated test suites catching real production bugs before release.
- **Auth microservice:** Expanded an assigned login module into a centralised authentication microservice for the entire organisation, serving **100+ users** across multiple applications — a decoupled, centralised service design.
- **Map component:** Designed an Angular + Leaflet map handling thousands of real-time assets.
- **Mentoring:** Onboarded an intern; built backend/frontend templates with built-in security and CI/CD.

`NestJS` `Node.js` `TypeScript` `Java` `Spring Boot` `PostgreSQL` `CouchDB` `Redis Stack` `Docker` `GitLab CI/CD` `Angular`

## Education

**MSc in Computer Science and Technology** — 2025 – 2026
*Universidad Carlos III de Madrid* — Madrid, Spain

**BSc in Software Engineering** — 2020 – 2025
*Universidad de Sevilla* — Seville, Spain

## Publications & Projects

*Multi-Label Emotion Detection in Early Modern Spanish Correspondence.* **D. Reyes Alés**, A. I. Arteaga, S. Peñafiel Calderón, A. Verdugo Valerio. HISEMOTIONS @ IberLEF 2026, CEUR-WS Vol. 4273. [PDF]

**req-classifier** (in progress) — Requirements classifier layering deterministic rules with Jev (TypeSafe AI), benchmarked against classical ML, a fine-tuned DeBERTa and Claude under a preregistered protocol. [GitHub]

## Technical Skills & Languages

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|---|---|
| **Backend** | Java · Spring Boot · NestJS · Node.js · REST APIs · Microservices |
| **Infra & Data** | PostgreSQL · MySQL · Redis Stack · CouchDB · Docker · GitLab CI/CD · Linux |
| **AI / ML** | Python · PyTorch · Hugging Face Transformers · scikit-learn · LLM APIs (Anthropic, Ollama) |
| **Other** | Angular · React/Next.js · JavaScript/TypeScript · Automated testing · Agile |
| **Languages** | Spanish (native) · English B2 First (FCE), Cambridge, 2023 |